Cricbuzz LiveStats

Real-Time Cricket Insights & SQL-Based Analytics

Abstract

This project presents a real-time cricket analytics dashboard built using Python and Streamlit. The application fetches live and recent cricket match data from RapidAPI and visualises it through an interactive user interface. The dashboard includes key performance indicators (KPIs), match analytics, and graphical representations using Plotly. The project demonstrates skills in API integration, data processing, visualisation, and deployment.

Introduction

Cricket analytics has become an essential part of modern sports analysis. This project aims to provide a centralised dashboard that displays real-time and historical cricket match insights.

The system is designed to:

Fetch live data using APIs

Process and structure the data

Display it in an interactive dashboard

Objectives

Build a real-time cricket dashboard

Integrate external APIs (RapidAPI)

Visualise match data effectively

Deploy the application on Streamlit Cloud

Ensure secure API key handling

Technologies Used

Python

Streamlit

Pandas

Plotly

Requests

RapidAPI

Git & GitHub

System Architecture

User Interface (Streamlit)
↓
Data Processing (Python)
↓
API Layer (RapidAPI)
↓
Live Cricket Data

Features of the System

Live & recent match tracking

KPI Metrics:

- Total Matches
- Live Matches
- Teams
- Venues

Data visualisation:

- Bar charts
- Pie charts

Multiple dashboard pages:

- Home
- Live

Players

Analytics

SQL

CRUD

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8. Implementation Details

The system fetches cricket data using REST API requests. The data is processed using Python and stored in a structured format. Streamlit is used to create an interactive dashboard interface, while Plotly is used for visualisations.

API keys are managed securely using environment variables and Streamlit secrets.

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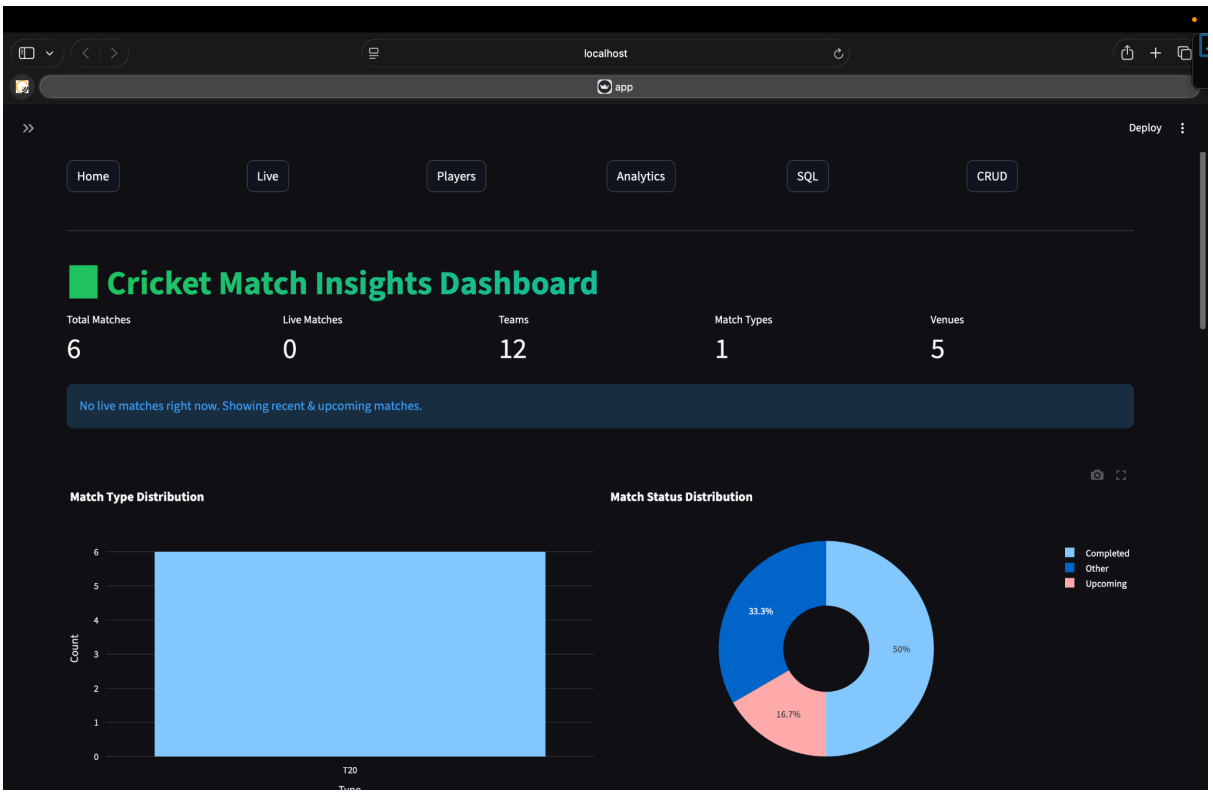
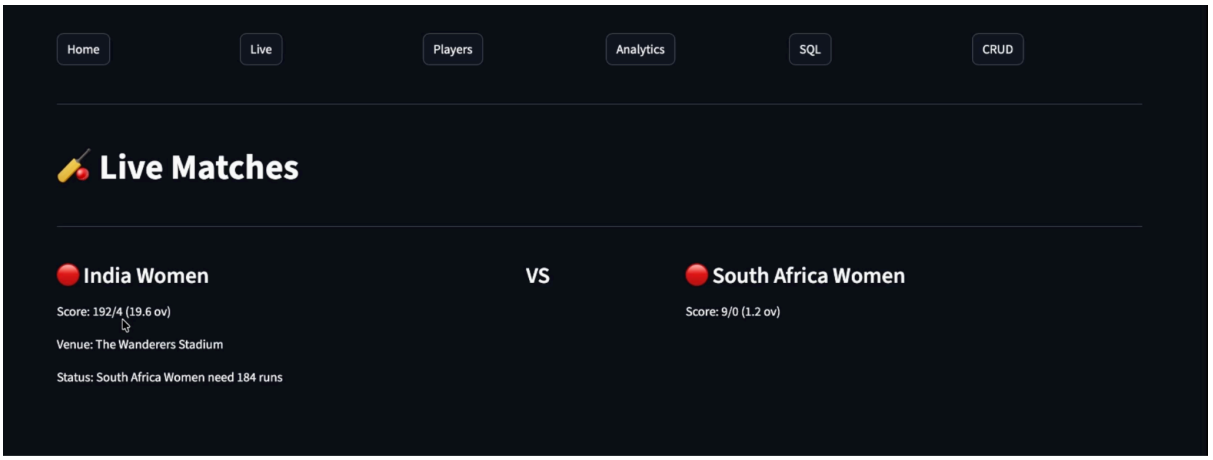
9. Screenshots

Screenshot 1: Home Dashboard



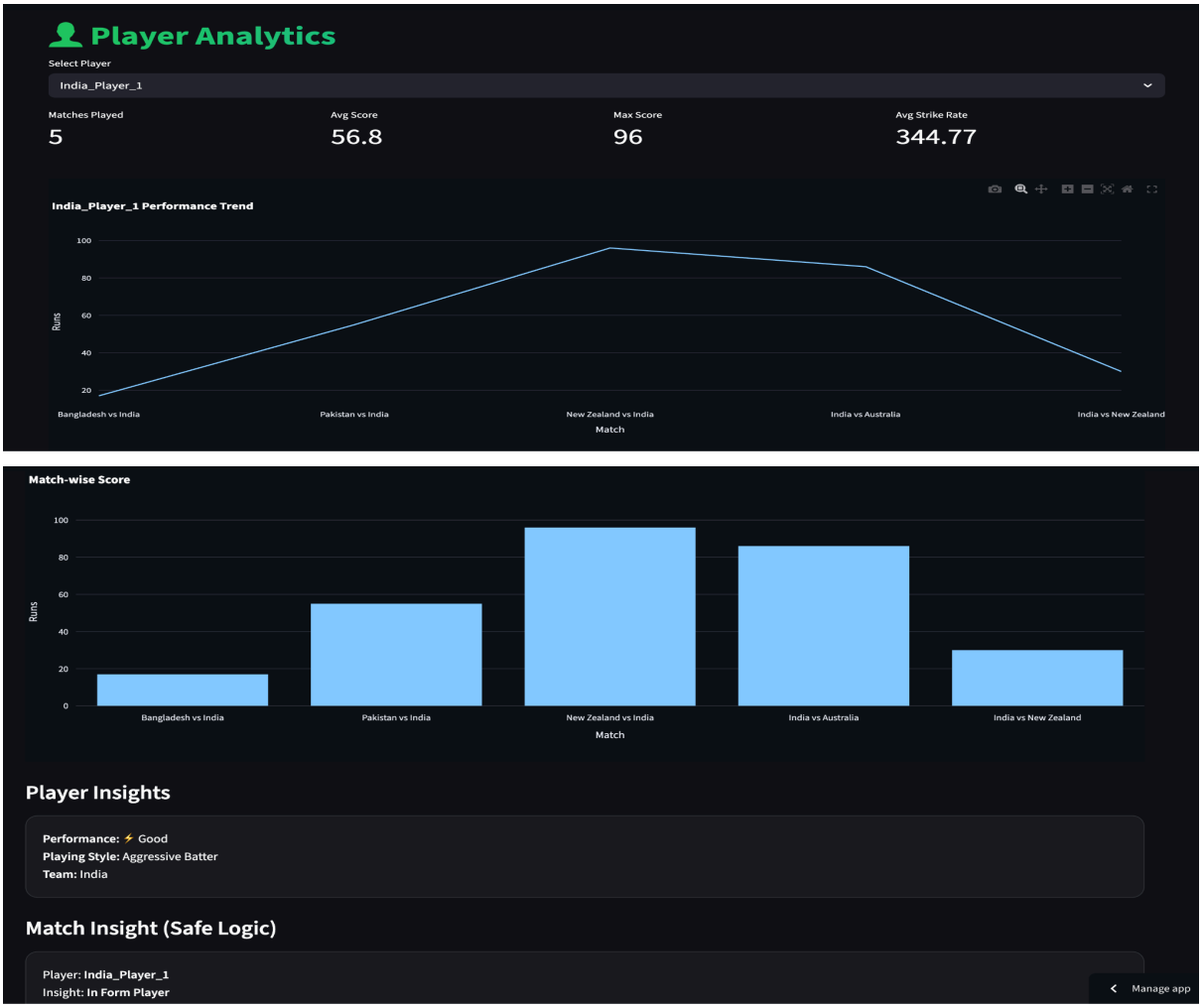
→ Shows KPIs and overview charts

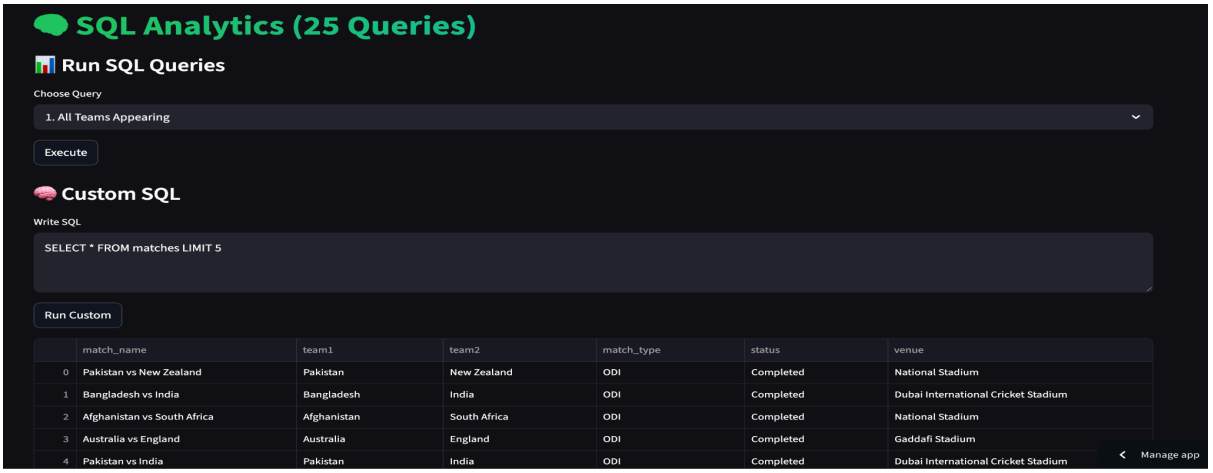
Screenshot 2: Live Matches Page

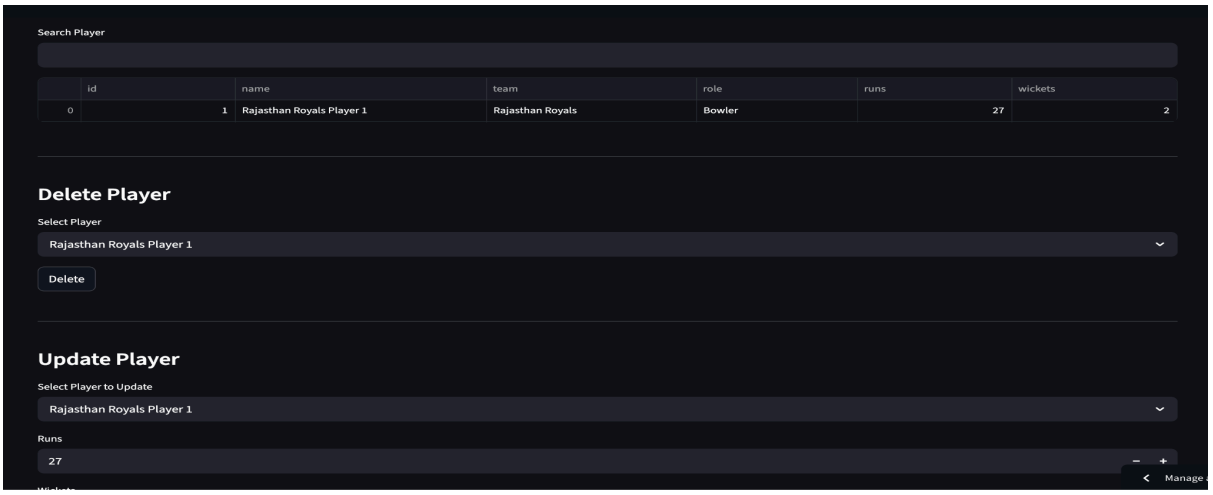


→ Displays ongoing matches

Screenshot 3: Analytics Page

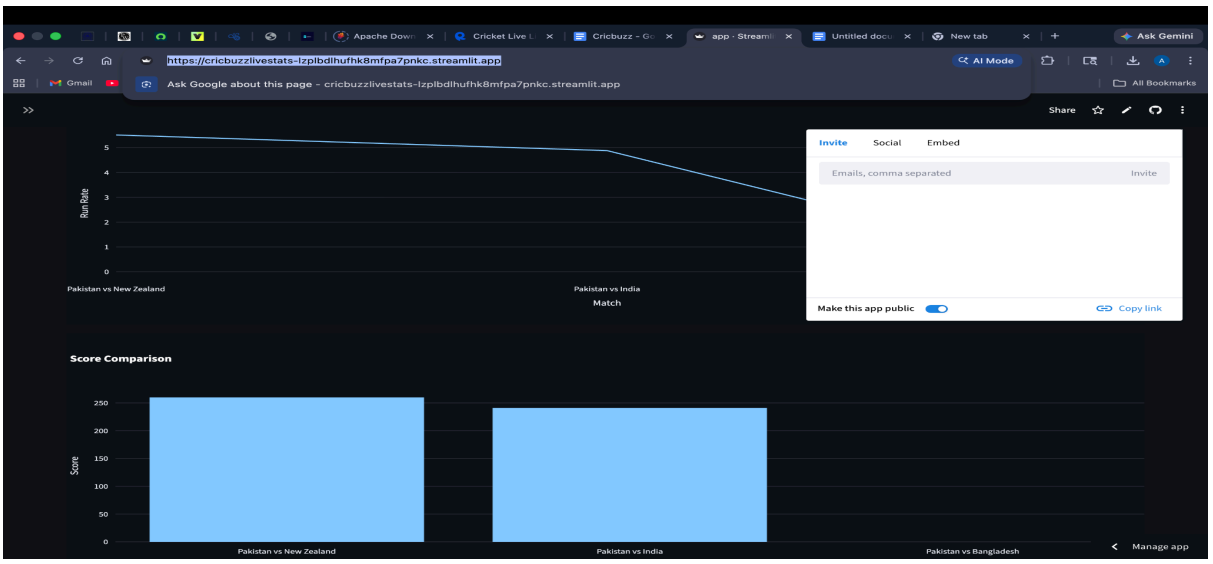






→ Shows graphs and distributions

Screenshot 4: Streamlit Deployment



→ Shows deployed app UI

Challenges Faced

API response inconsistencies

Deployment errors on Streamlit Cloud

Secure handling of API keys

Git merge and push conflicts

Solutions

Used error handling for API responses

Added requirements.txt for dependencies

Used .env locally and Streamlit secrets in deployment

Resolved Git issues using rebase and force push

Deployment

The project is deployed using Streamlit Cloud.

Steps:

Push code to GitHub

Connect the repository to Streamlit

Add API key in secrets

Deploy application

Results

The dashboard successfully displays real-time cricket data with interactive visualisations. The system is responsive and provides meaningful insights into match statistics.

Future Enhancements

Player-level analytics

Match prediction system

AI-based insights

Advanced filtering options

Mobile-friendly UI

Conclusion

This project demonstrates the practical implementation of data analytics using Python. It highlights the integration of APIs, data visualisation, and deployment. The dashboard provides a strong foundation for building advanced sports analytics systems.

References

https://streamlit.io

https://rapidapi.com

GitHub & Live Links

GitHub Repository:
https://github.com/AchintyaKrishna/Cricbuzz_LiveStats

Live App:
https://cricbuzzlivestats-lzplbdlhufhk8mfpa7pnkc.streamlit.app/

YouTube Demo:
https://youtu.be/9SQNvkZtwzg?si=g6mk4j13DHWi3mXQ

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